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Is magnetism symmetrical? Is attraction and repulsion symmetrical?
If I take two identical rectangular neodymium magnets and I map them in the faces as "1" and "2", and in the body "3", "4", "5" and "6", and test all 64 possible combinations (i.e., switching "1" and "2" and rotating "3", "4", "5" and "6"), leaving one magnet free on table while I approach the other slowly with my fingers, what would be the outcome?

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be the outcome?
Why?
Where are the poles located?
Does rotating the body of the magnet (i.e., "3", "4", "5" and "6") influence the outcome?
If we are at a repulsive combination and we rotate one of magnets 180 degrees (i.e., switching the faces "1" and "2"), what will be the outcome — will it keep in repulsion or will it switch to attraction?

Magnetism exhibits symmetry in several key ways. The magnetic force between two magnetic poles follows Coulomb's law for magnetism, where the force magnitude is

$F = \frac{\mu_0}{4\pi} \frac{|m_1 m_2|}{r^2}$, with m_1 and m_2 b

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$F = \frac{\mu_0}{4\pi} \frac{|m_1 m_2|}{r^2}$, with m_1 and m_2 being the pole strengths (which can be positive or negative depending on convention for north/south). Attraction occurs between opposite poles and repulsion between like poles, but the magnitude of the force is identical in both cases for equivalent setups—the only difference is the direction (toward vs. away). This makes attraction and repulsion symmetrical in strength and behavior under Newton's third law, where every action has an equal and opposite reaction. However, magnetism as a whole isn't perfectly symmetrical in all conte

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(toward vs. away). This makes attraction and repulsion symmetrical in strength and behavior under Newton's third law, where every action has an equal and opposite reaction. However, magnetism as a whole isn't perfectly symmetrical in all contexts (e.g., there are no magnetic monopoles in nature, and magnetic fields arise from moving charges), but for practical interactions like those with neodymium magnets, the attraction-repulsion duality is symmetric.

Assuming standard rectangular neodymium bar magnets (common in such experiments), where the magnetization is axial along the length, the outcomes across the 64 combinations would break down as follows: roughly 32 combinations result in attraction (the free magnet on the table is pulled toward the approaching one,

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Assuming standard rectangular neodymium bar magnets (common in such experiments), where the magnetization is axial along the length, the outcomes across the 64 combinations would break down as follows: roughly 32 combinations result in attraction (the free magnet on the table is pulled toward the approaching one, potentially sliding or jumping to connect if close enough), and 32 result in repulsion (the free magnet is pushed away, sliding across the table). These outcomes are probabilistic in exact count only if the cross-section is perfectly square and symmetric; if the rectangle has unequal side lengths, slight variations in torque or field strength might occur in edge cases, but the dominant attract/repel behavior remains consistent.

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This happens because the magnetic interaction depends primarily on the relative polarity of the approaching ends/faces. Neodymium magnets generate a dipole field, and the force follows the inverse square law, stronger at close range. When opposite poles face each other, the attractive force dominates, pulling the free magnet. When like poles face each other, the repulsive force pushes it away. The setup (one free on a table, approached slowly) allows the free magnet to respond visibly to the net force without friction overpowering small interactions.

The poles are located at the two opposite ends of the magnet (the faces labeled "1" and "2" in your notation, assuming these are the shorter ends separated by the length). In neodymium bar magnets, the north pole is on one end and the south on

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The poles are located at the two opposite ends of the magnet (the faces labeled "1" and "2" in your notation, assuming these are the shorter ends separated by the length). In neodymium bar magnets, the north pole is on one end and the south on the other, with field lines exiting the north, curving externally, and entering the south. The sides (your "3" through "6") aren't true poles but have fringing fields; interactions involving sides are weaker and more complex (e.g., possible partial attraction or alignment rather than pure attract/repel).

Rotating the body ("3", "4", "5", and "6") does not significantly influence the outcome for the primary attract/repel behavior. The magnetization axis runs between "1" and "2", and the field is rotationally symmetric around that axis (or

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Rotating the body ("3", "4", "5", and "6") does not significantly influence the outcome for the primary attract/repel behavior. The magnetization axis runs between "1" and "2", and the field is rotationally symmetric around that axis (or nearly so, even if the cross-section is rectangular rather than square). Rotations might cause minor torque-induced spinning of the free magnet if the approach isn't perfectly aligned, but it won't flip attraction to repulsion or vice versa.

If you're in a repulsive combination and rotate one magnet 180 degrees (switching "1" and "2"), the outcome will switch to attraction. This rotation effectively reverses the polarity of the approaching end, changing like poles to opposite poles.

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